

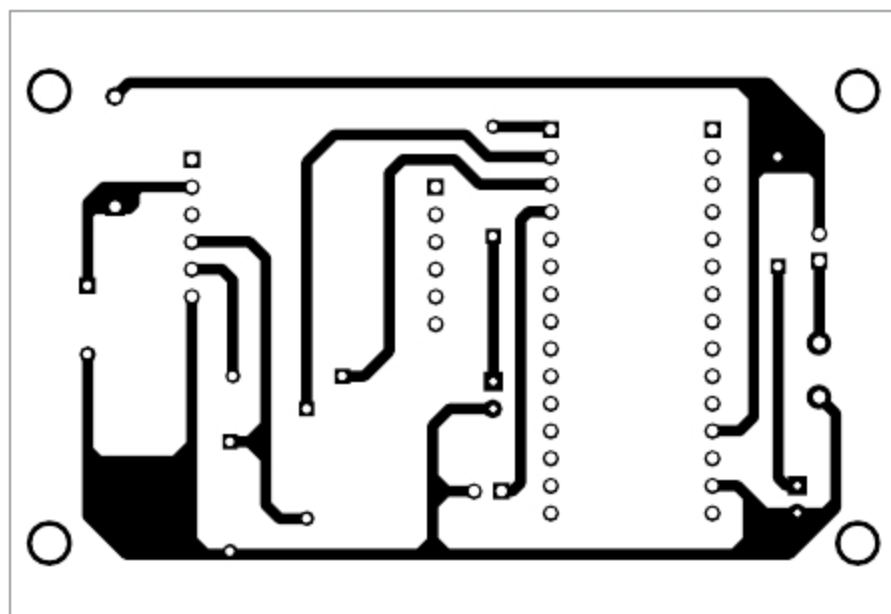


Fig. 4: Actual-size PCB layout of the GSM security switch

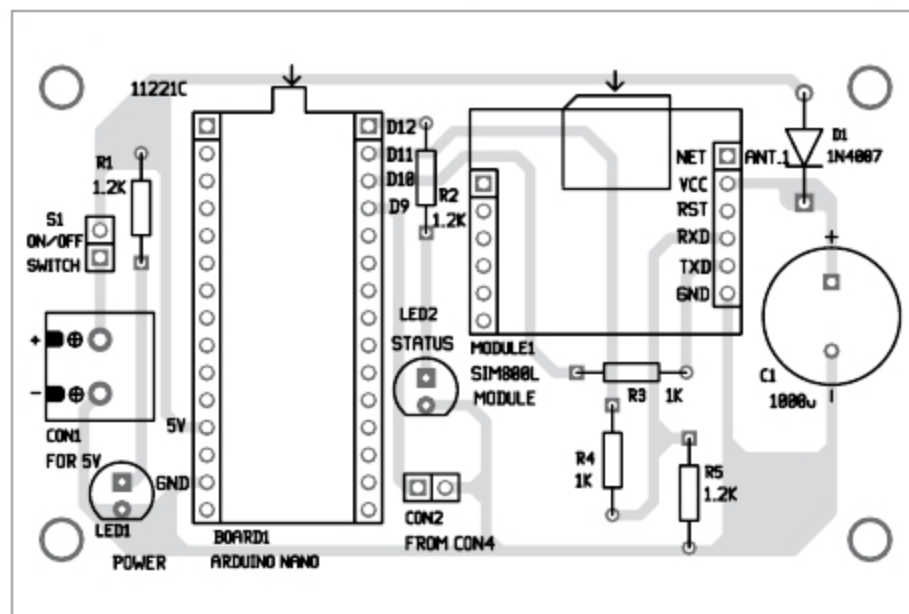


Fig. 5: Components layout for the PCB shown in Fig. 4

PARTS LIST

Semiconductors:

Board1	- Arduino Nano
Module1	- SIM800L module
IC1	- NE555 timer
T1	- BC547 npn transistor
D1	- 1N4007 rectifier diode
D2	- 1N4148 signal diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2, R5	- 1.2-kilo-ohm
R3, R4	- 1-kilo-ohm
R6	- 100-kilo-ohm
R10	- 10-kilo-ohm

Capacitors:

C1	- 1000 μ F, 16V electrolytic
C2	- 10 μ F, 16V electrolytic
C3, C4	- 100nF ceramic disk

Miscellaneous:

TS1	- Tilt switch or touch plate
CON1, CON3	- 2-pin terminal connector
CON2, CON4	- 2-pin connector
S1	- On/off switch
	- Valid activated SIM
	- Micro-USB cable for Arduino
	- PCBs

(All these are available from kitsnspares.com)

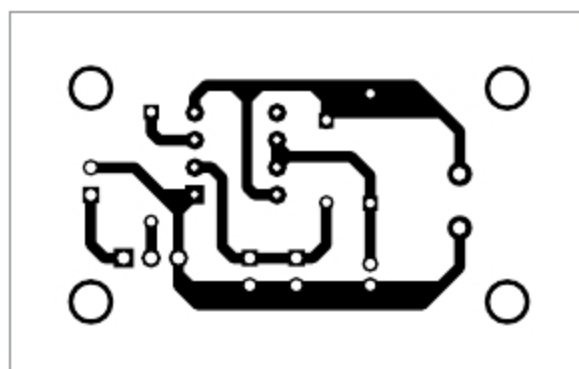


Fig. 6: Actual-size PCB layout of the touch/tilt sensor for vehicle anti-theft alert

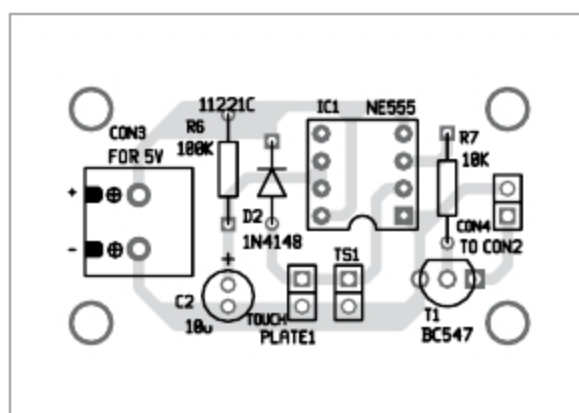


Fig. 7: Components layout for the PCB shown in Fig. 6

and its components layout in Fig. 5. Upload the source code to Arduino Nano and then insert it into the PCB. Solder the spiral antenna on SIM800L module.

Mount Arduino Nano and SIM800L on the PCB. Connect 5V, 2A regulated power supply across CON1. When security switch connected to pin D9 of Arduino Uno goes low momentarily, an alert SMS carrying 'Warning, Intrusion' message will be received on the cellphone.

An actual-size PCB layout of the tilt/touch sensor circuit is shown in Fig. 6 and its components layout in Fig. 7. Assemble the circuit on the PCB and fix either the touch plate or tilt sensor or both, as shown in the circuit diagram of Fig. 3. Connect 5V DC regulated power supply across CON3. Then, connect CON2 to sensor circuit CON4 using a two-wire cable.

On touching sensor plate1 or tilting sensor TS1, output of IC1 goes high and transistor T1 conducts. This, in turn, pulls pin D9 of Arduino to ground through T1. This action triggers GSM switch and generates an SMS alert message, which is received on the cellphone.

The project is now ready to use. Place the circuit at a suitable location to protect the asset (like electronic safe, vehicle, etc) or property. **EFY**

phone number (phone number for receiving the alert message).

Interfacing with the sensor

You can connect sensor input at CON2 with such sensors as reed switches, photo-resistors/diodes and thermostats, allowing for considerable customisation of the security system. This also adds a number of capabilities including recognition and response to momentary inputs from a laser trip wire, impact sensor, door/gate sensor, etc.

An optional sensor front-end model for the security system using an inexpensive touch/tilt sensor circuit is

described here, as shown in Fig. 3. It is built around a touch plate, tilt switch (TS1), NE555 timer (IC1), BC547 npn transistor (T1), 5V power supply and a few other components.

Construction and testing

An actual-size PCB layout of the GSM security switch is shown in Fig. 4

EFY Note

The source code of this project is available for free download at source.efymag.com



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